

Exercise 3	Github+Conflict+Check
20/7/26	

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Task 1 – Repository Setup

```
C:\Users\iot-b2\lab3>git init
Initialized empty Git repository in C:/Users/iot-b2/lab3/.git/

C:\Users\iot-b2\lab3>mkdir one

C:\Users\iot-b2\lab3>mkdir two

C:\Users\iot-b2\lab3>mkdir three

C:\Users\iot-b2\lab3>git add .

C:\Users\iot-b2\lab3>git commit -m"first"
[master (root-commit) 142cb0c] first
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

C:\Users\iot-b2\lab3>git branch -M main

C:\Users\iot-b2\lab3>git remote add origin https://github.com/Tharun-Luffy/lab-20-07-.git

C:\Users\iot-b2\lab3>git push origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 232 bytes | 232.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Tharun-Luffy/lab-20-07-.git
 * [new branch]      main -> main
```

```
C:\Users\iot-b2\lab3>git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    one/
    three/
    two/

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\iot-b2\lab3>git add .

C:\Users\iot-b2\lab3>git commit -m"second"
[main 3291330] second
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 one/test.txt
 create mode 100644 three/a.txt
 create mode 100644 two/b.txt

C:\Users\iot-b2\lab3>git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 463 bytes | 463.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Tharun-Luffy/lab-20-07-.git
 142cb0c..3291330  main -> main

C:\Users\iot-b2\lab3>
```

Task 2 – Clone of Repo and creation of new modules

```
C:\Users\iot-b2\lab3\src>type nul > register.py

C:\Users\iot-b2\lab3\src>open register.py
'open' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\iot-b2\lab3\src>notepad register.py

C:\Users\iot-b2\lab3\src>git add .

C:\Users\iot-b2\lab3\src>git commit -m"feat:create register"
[main 3d65eb6] feat:create register
 1 file changed, 2 insertions(+)
 create mode 100644 src/register.py

C:\Users\iot-b2\lab3\src>git push
fatal: The current branch main has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin main

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

C:\Users\iot-b2\lab3\src>git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 385 bytes | 385.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Tharun-Luffy/lab-20-07-.git
 46cd397..3d65eb6  main -> main
branch 'main' set up to track 'origin/main'.
```

```
C:\Users\iot-b2\Downloads\sar-git>git init
Initialized empty Git repository in C:/Users/iot-b2/Downloads/sar-git/.git/

C:\Users\iot-b2\Downloads\sar-git>git pull https://github.com/Tharun-Luffy/lab-20-07-.git
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 9 (delta 0), reused 9 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (9/9), 643 bytes | 13.00 KiB/s, done.
From https://github.com/Tharun-Luffy/lab-20-07-
 * branch                HEAD              -> FETCH_HEAD
```

```

C:\Users\iot-b2\Downloads\sar-git>type nul > login.py

C:\Users\iot-b2\Downloads\sar-git>git add .

C:\Users\iot-b2\Downloads\sar-git>git commit -m"sar commit"
[master b6913e2] sar commit
1 file changed, 2 insertions(+)
create mode 100644 src/login.py

C:\Users\iot-b2\Downloads\sar-git>git remote add origin https://github.com/Tharun-Luffy/lab-20-07-.git

C:\Users\iot-b2\Downloads\sar-git>git remote -v
origin https://github.com/Tharun-Luffy/lab-20-07-.git (fetch)
origin https://github.com/Tharun-Luffy/lab-20-07-.git (push)

C:\Users\iot-b2\Downloads\sar-git>git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 435 bytes | 435.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote: https://github.com/Tharun-Luffy/lab-20-07-/pull/new/master
remote:
To https://github.com/Tharun-Luffy/lab-20-07-.git
 * [new branch] master -> master
branch 'master' set up to track 'origin/master'.

C:\Users\iot-b2\Downloads\sar-git>git push -u origin master:main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Tharun-Luffy/lab-20-07-.git
 3291330..b6913e2 master -> main
branch 'master' set up to track 'origin/main'.

```

Task 3 – Merge Conflict

```

C:\Users\iot-b2\Downloads\sar-git>git push -u origin main
To https://github.com/Tharun-Luffy/lab-20-07-.git
 ! [rejected] main -> main (fetch first)
error: failed to push some refs to 'https://github.com/Tharun-Luffy/lab-20-07-.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\Users\iot-b2\Downloads\sar-git>git pull origin main
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 11 (delta 2), reused 8 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (11/11), 112.56 KiB | 1.66 MiB/s, done.
From https://github.com/Tharun-Luffy/lab-20-07-
 * branch main -> FETCH_HEAD
 b6913e2..d40b3cb main -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

```

```

<<<<<< HEAD
"# lab-20-07- modified(conflict)"
=====
"# lab-20-07-26"
>>>>>> d40b3cba82ffc3749876a045e90e98d118344198
|

```

Task 4 – Deleting someone's file

```

C:\Users\iot-b2\Downloads\sar-git>git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 1), reused 4 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 371 bytes | 15.00 KiB/s, done.
From https://github.com/Tharun-Luffy/lab-20-07-
 d40b3cb..a88e4e main -> origin/main
CONFLICT (modify/delete): src/register.py deleted in HEAD and modified in a88e4e4ef40397a0a0328d1fb795941bd8c3b773a. Version a88e4e4ef40397a0a0328d1fb795941bd8c3b773a of src/register.py left in tree.
Automatic merge failed; fix conflicts and then commit the result.

```

Task 5 – Same function Conflict

```
C:\Users\iot-b2\Downloads\sar-git\src>notepad calculator.py
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git add .
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git commit -m "added calculation"
[main 21f0804] added calculation
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git push
Enumerating objects: 19, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 20 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 1.11 KiB | 568.00 KiB/s, done.
Total 10 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/Tharun-Luffy/lab-20-07-.git
a88e44e..21f0804 main -> main
```

```
C:\Users\iot-b2\lab3\src>notepad calculator.py
```

```
C:\Users\iot-b2\lab3\src>git add .
```

```
C:\Users\iot-b2\Downloads\sar-git\src>notepad calculator.py
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git add calculator.py
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git commit -m "Changed calculation to SI"
[main dd48805] Changed calculation to SI
1 file changed, 1 insertion(+), 1 deletion(-)
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git push
To https://github.com/Tharun-Luffy/lab-20-07-.git
! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/Tharun-Luffy/lab-20-07-.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 1), reused 4 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 400 bytes | 19.00 KiB/s, done.
From https://github.com/Tharun-Luffy/lab-20-07-
21f0804..a6489a8 main -> origin/main
Auto-merging src/calculator.py
CONFLICT (content): Merge conflict in src/calculator.py
Automatic merge failed; fix conflicts and then commit the result.
```

```
C:\Users\iot-b2\Downloads\sar-git\src>notepad calculator.py
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git push
To https://github.com/Tharun-Luffy/lab-20-07-.git
! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/Tharun-Luffy/lab-20-07-.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git add .
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git commit -m "changed/resolved calculation"
[main eec9a63] changed/resolved calculation
```

```
C:\Users\iot-b2\Downloads\sar-git\src>git push
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
```

Task 6: Cherry Pick

```
C:\TS\personal\repos\devops\lab-20-07->git checkout b1
Switched to branch 'b1'
```

```
C:\TS\personal\repos\devops\lab-20-07->cd src
```

```
C:\TS\personal\repos\devops\lab-20-07-\src>cd src & notepad cherry-report.py
The system cannot find the path specified.
```

```
C:\TS\personal\repos\devops\lab-20-07-\src>git add .
```

```
C:\TS\personal\repos\devops\lab-20-07->git commit -m "cherry-commit-1"
[b1 2e626d1] cherry-commit-1
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 src/cherry-report.py
```

```
C:\TS\personal\repos\devops\lab-20-07->git push origin b1
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 408 bytes | 408.00 KiB/s, done.
Total 4 (delta 1), reused 1 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Tharun-Luffy/lab-20-07-.git
    50d94a3..2e626d1  b1 -> b1
```

```
C:\TS\personal\repos\devops\lab-20-07->git checkout b2
A      src/cherry-calculator.py
Switched to branch 'b2'
```

```
C:\TS\personal\repos\devops\lab-20-07->cd src & notepad cherry-calculator.py
```

```
C:\TS\personal\repos\devops\lab-20-07-\src>git add .
```

```
C:\TS\personal\repos\devops\lab-20-07-\src>git commit -m "cherry-commit-2"
[b2 77fa85c] cherry-commit-2
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 src/cherry-calculator.py
```

```
C:\TS\personal\repos\devops\lab-20-07->git push origin b2
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 401 bytes | 401.00 KiB/s, done.
Total 4 (delta 1), reused 1 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'b2' on GitHub by visiting:
remote:      https://github.com/Tharun-Luffy/lab-20-07-/pull/new/b2
remote:
To https://github.com/Tharun-Luffy/lab-20-07-.git
 * [new branch]      b2 -> b2
```

```

C:\TS\personal\repos\devops\lab-20-07->git log --oneline
77fa85c (HEAD -> b2, origin/b2) cherry-commit-2
50d94a3 (origin/main, origin/HEAD) Add files via upload
fbb7c9f images
eec9a63 changed/resolved calculation
a6489a8 feat:modified calculator.py
dd48805 Changed calculation to SI
21f0804 added calculation
a88e44e feat:change in register
de170d4 deleted register.py
d40b3cb changed readme
dc5cdbe "sar modified readme"
3d65eb6 feat:create register
46cd397 images
b6913e2 (origin/master) sar commit
3291330 second
142cb0c first

C:\TS\personal\repos\devops\lab-20-07->git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)

C:\TS\personal\repos\devops\lab-20-07->git cherry-pick 77fa85c
[main 89a03de] cherry-commit-2
Date: Mon Jul 20 14:41:59 2026 +0530
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 src/cherry-calculator.py

C:\TS\personal\repos\devops\lab-20-07->git log --oneline
89a03de (HEAD -> main) cherry-commit-2
1e733d0 Revert "cherry-commit-1"
269f292 cherry-commit-1
50d94a3 (origin/main, origin/HEAD) Add files via upload
fbb7c9f images

```

Task 7 : Revert

```

C:\TS\personal\repos\devops\lab-20-07->git add .

C:\TS\personal\repos\devops\lab-20-07->git commit -m "acvidential deletition login.py"
[main 3c37eea] acvidential deletition login.py
1 file changed, 2 deletions(-)
delete mode 100644 src/login.py

C:\TS\personal\repos\devops\lab-20-07->git log --oneline
3c37eea (HEAD -> main) acvidential deletition login.py
89a03de cherry-commit-2
1e733d0 Revert "cherry-commit-1"
269f292 cherry-commit-1
50d94a3 (origin/main, origin/HEAD) Add files via upload
fbb7c9f images
eec9a63 changed/resolved calculation
a6489a8 feat:modified calculator.py
dd48805 Changed calculation to SI
21f0804 added calculation
a88e44e feat:change in register
de170d4 deleted register.py
d40b3cb changed readme
dc5cdbe "sar modified readme"
3d65eb6 feat:create register
46cd397 images
b6913e2 (origin/master) sar commit
3291330 second
142cb0c first

C:\TS\personal\repos\devops\lab-20-07->git revert 3c37eea
[main 9a93f2b] Revert "acvidential deletition login.py"
1 file changed, 2 insertions(+)
create mode 100644 src/login.py

```